

Productive-flow share and wealth-coupled cultural transmission: a stylised agent-based model of cultural erosion under economic pressure

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Abstract

We present an agent-based model of an economy with seven occupationally heterogeneous agent classes that exchanges its monetary state through production, extraction, credit, taxation, inheritance, and a bandwidth-limited cultural-transmission channel. The cultural channel couples to wealth on a log scale: the number of skill dimensions a child inherits from a parent rises as the log of parent wealth, so the transmission bandwidth is graded across orders of magnitude of wealth rather than saturating at a single threshold.

We argue and demonstrate three results that we believe make the model useful for an artificial-life audience.

First, the model is honest about its scope. The $K=5$ skill space is *closed* by construction: mutation can reset a dimension to a fresh draw but no genuinely new skill dimension can appear. The model therefore exhibits cultural *erosion* under wealth concentration rather than cumulative cultural evolution. This is the right framing relative to Henrich [2015] and Henrich [2004], with Vaesen et al. [2016] noted as a counter-case. We present a minimal open-ended extension that allows new skill dimensions to enter the global pool via wealth-biased novelty events, and show on a per-capita basis that the discovery channel concentrates novelty in wealthier lineages.

Second, the cultural-economic asymmetry of the model, in which cultural transmission alone does not equalise wealth even when equalised at birth, is grounded in mechanism rather than in architectural wiring. A class-by-class regression of 40-tick wealth gain on the skill vector shows that workers and makers have strong skill-to-wealth fits ($R^2 \geq 0.97$ with normalised elasticities $\eta \geq 0.76$ on the class-typical dimensions); rentiers, lenders, and speculators have near-zero fits ($R^2 \leq 0.13$, $|\eta| \leq 0.10$). The cultural-economic asymmetry is therefore the consequence of one half of the economy (extractive classes) having weak internal skill-wealth feedback while the other half (productive classes) has strong feedback.

Third, we introduce a continuous per-tick scalar, the *productive-flow share* $\Phi_t = F_t^{\text{prod}} / (F_t^{\text{prod}} + F_t^{\text{ext}})$, which captures whether per-tick monetary accrual is dominated by productive- or extractive-class activity. Φ is not an autocatalytic-set closure in the Kauffman [1988] / Hordijk and Steel [2004] RAF sense; it is a flow-share measure operationalising the productive-vs-extractive distinction that the heterodox economic tradition has long named without instrumenting. The Sankey of steady-state per-tick flows makes Φ legible as a diagram rather than as a scalar.

We note in passing a result from the policy-comparison side of the broader project: an absolute cap on inheritance raises the wealth share held by the bottom 50% of the population from 0.0% under status quo to 19.2% (cap alone) or 25.3% (cap plus 2% wealth tax). The same intervention raises the productive-flow share Φ from approximately zero to 0.27–0.37. Among the instruments we tested, the cap is the only one that changes both the ownership distribution and the productive–extractive flow balance materially. The cultural-economic asymmetry result of this paper provides part of the mechanistic explanation: the extractive classes have weak skill-to-wealth feedback, so redistributing skills does not redistribute wealth, but redistributing the inheritance of wealth itself does. The policy-discourse implications of the cap result are reported in the companion submission targeting Fiscal Studies.

We position these cultural contributions relative to the cultural-evolution and open-ended-evolution literatures, locate the model relative to Spangler and Sarkar [2022] as the closest published economic-ABM neighbour, and discuss the limits of the cultural model and the open-skills extension. The fiscal-policy results that follow naturally from the model are summarised in a companion submission targeting the heterodox-economics literature.

1 Introduction

The artificial-life community has long studied open-ended evolution, cumulative cultural transmission, and the autocatalytic structure of chemical and biological systems. This paper presents a model that borrows from each of those traditions but applies them to a question the alife literature has not previously taken on directly: how economic inequality and cultural diversity co-evolve, and what mechanisms cause the cultural sphere to track the economic sphere.

We approach this with a stock-flow agent-based simulation. Seven occupationally typed agents (workers, makers, employers, lenders, rentiers, speculators, and a single state) exchange a monetary state variable through production, extraction (rent, interest, capital return), inheritance, and a bandwidth-limited cultural-transmission channel. Each agent carries a five-dimensional skill vector; on death a same-class child inherits a number of skill dimensions that scales with the log of parent wealth. The mechanism is simple, and it has the property that cultural inheritance concentrates in wealthier lineages not by an explicit class bias but by the wealth-coupling of the bandwidth gate itself.

We organise the paper around three contributions.

The first is honesty about scope. The $K=5$ skill space is closed by construction. In this respect the model is squarely on one side of the open-endedness debate [Taylor, 2019, Banzhaf et al., 2016]: the only thing that can evolve in the main model is the wealth distribution and the subset of the existing skill pool held by each agent. The Henrich [2015] “cumulative culture” framing therefore does not strictly apply. What the model demonstrates is the *erosion* of cultural breadth under wealth concentration, in the spirit of Henrich [2004], with Vaesen et al. [2016]’s critique of the Tasmanian inference noted. We add (§7) a stand-alone open-ended extension in which new skill dimensions can enter the pool via wealth-biased novelty events.

The second is the mechanism behind the cultural-economic asymmetry. The model exhibits a striking negative result: cultural transmission alone does not equalise wealth, even when transmission is equalised at birth. An obvious worry is that this asymmetry is built into the architecture. We address that worry by quantifying the skill-to-wealth channel directly with a class-by-class regression of wealth gain on the skill vector. The asymmetry turns out to be partly structural: the extractive classes (rentiers, lenders, speculators) have R^2 between 0.11 and 0.13 and normalised elasticities of order 0.05 to 0.10. But the productive classes (workers, makers) have R^2 of 0.97 to 0.98 and normalised elasticities of 0.76 to 0.95 on the class-typical dimensions. The asymmetry is therefore not an arbitrary wiring choice but the consequence of one half of the economy having strong internal skill-to-wealth feedback and the other half not. The model is doing the right thing.

The third is a measurement device. We define Φ_t , the productive-flow share, as the share of per-tick monetary accrual accruing to or being created by productive-class agents: $\Phi_t = F_t^{\text{prod}} / (F_t^{\text{prod}} + F_t^{\text{ext}})$. We use this to characterise regimes (Section 4) and to visualise wealth flows (the Sankey, Section 5). We explicitly distinguish Φ from the topological autocatalytic-set closure of Kauffman [1988] and Hordijk and Steel [2004], after which it is named in motivation but not in technical content.

The fiscal-policy implications of the model (wealth-tax design, inheritance trapdoor, calibrated-vs-default policy comparisons) are substantial and are reported separately in a companion submission; this paper engages them only to the extent needed to give a cultural- evolutionary reading of the results.

2 Related work

The most direct alife-adjacent precedent on cultural transmission is the cultural-evolution tradition of Cavalli-Sforza and Feldman [1981], Boyd and Richerson [1985], Henrich [2015], and Lewis and Laland [2012]. The wealth-coupled bandwidth mechanism we introduce sits adjacent to: Henrich and Gil-White [2001] on prestige-biased transmission (where the *target* of imitation is status-biased; we couple the *bandwidth* of transmission to wealth instead); Henrich [2004] on population-size-capped cumulative culture, where Vaesen et al. [2016] provide the counter-analysis arguing the relevant population sizes in the Tasmania case were not as different as the model required; Lewis and Laland [2012] on the empirical sensitivity of cumulative culture to transmission fidelity; and Charbonneau and Bourrat [2021] on the “grain problem” for fidelity measures, which directly applies to our $K=5$ discretisation. We have not been able to locate a published agent-based model that couples transmission *bandwidth* to wealth specifically.

On the alife/OEE side we engage with Hashimoto et al. [2019] on open-ended evolution of tag combinations, Taylor [2019] on routes to open-ended evolution, and Banzhaf et al. [2016] on the OEE program more generally. The TAP-style adjacent-possible framing of Hordijk and Kauffman [2023] motivates our open-ended skill-discovery extension.

The closest economic-ABM neighbour is Spangler and Sarkar [2022], who couple estate taxation with binary skill inheritance and a Carnegie effect. We test their headline mechanism directly under a clean toggle in §6.4.

For the productive-vs-extractive framing of Φ we draw on Kauffman [1988] and Mazzucato [2018]; Hordijk and Kauffman [2023] is the closest neighbour on detecting autocatalytic structure in production-function networks, where they use a topological RAF detection and we use a continuous flow ratio.

3 The model

3.1 Agent types and population

Seven agent classes are present at fixed proportions per 100 agents: 72 workers, 9 makers, 7 employers, 5 lenders, 3 rentiers, 3 speculators, and 1 state. Each agent carries wealth, debt, age, lifespan drawn from $\mathcal{N}(70, 12)$ truncated to $[20, 120]$, a five-dimensional skill vector with class-typical biases at initialisation, peak lifetime wealth, and a bandwidth-at-birth record used for the analysis in §6.2.

The skill vector dimensions are interpreted as *technical*, *organisational*, *social*, *financial*, and *generative*. Initialisation gives each class a class-typical prior bias on a subset of dimensions; the bias is small and is preserved by the inheritance rule below.

The model runs at $N = 100$ in the cultural results reported here. A scale-up to $N = 10,000$ is available; the cultural results are qualitatively unchanged under scaling and we report them at $N = 100$ for legibility.

3.2 Per-tick mechanics, in brief

Each tick runs the following phases in order. Production: makers and employers hire workers from a skill-sorted queue; makers transform a material input plus paid labour into output at multiplier M ; employers capture a markup. State optionally hires unemployed workers and purchases maker output. Extraction: workers pay rent to rentiers in proportion to rentier wealth; rentiers earn capital return r ; speculators accrue a stochastic intra-stock return. Consumption and credit: subsistence is debited from every non-state agent; shortfalls are borrowed from a lender pool, with starvation if the pool is exhausted. Interest and repayment: debtors pay interest when liquid, principal when wealth allows. Tax and UBI. Life: agents whose age exceeds their lifespan die naturally; a child of the same class is born with inherited wealth

(after inheritance tax and cap) and bandwidth-limited inherited skills (next subsection). A complete ODD-style summary is in Appendix A.

3.3 Cultural transmission: bandwidth-limited inheritance

When a parent agent of class t dies and is replaced by a same-class child, the child’s skill vector is constructed as follows. The bandwidth, i.e. the number of skill dimensions the child inherits from the parent, is

$$\text{bw_dims}(w_{\text{parent}}) = \text{clip}\left(b_0 K + b_w K \cdot \log_{10}(\max(1, w_{\text{parent}}))/4, 1, K\right), \quad (1)$$

with $K = 5$, b_0 a base proportion ($b_0 = 0.5$ default), and b_w a wealth-coupling slope ($b_w = 0.5$ default). The child inherits the parent’s bw_dims strongest skill dimensions; the rest are zero. A small per-dimension mutation rate ($\mu = 0.05$ default) replaces an inherited value with a fresh Beta(2,2) draw on its own account.

The log-wealth coupling is the key mechanism for this paper. With parent wealth at 1 unit, $\log_{10}(1) = 0$, so $\text{bw_dims} = b_0 K = 2.5$ (rounded to 3). At parent wealth 10,000 units, $\log_{10}(10000)/4 = 1$, so $\text{bw_dims} = b_0 K + b_w K = 5$ (all skills transmitted). The coupling is thus graded across four orders of magnitude of parent wealth, rather than saturating at a single intermediate threshold as a prior linear formulation did.

Knowledge loss is recorded as the per-tick sum of skill mass present in each dying parent but not transmitted to its child.

3.4 The productive-flow share metric

During each tick we accumulate two flow sums treated symmetrically. Productive accrual F_t^{prod} sums (i) wages paid by makers, employers, and the state to workers; (ii) state purchases of maker output; (iii) maker value-add, i.e. the surplus output – inputs created in production and retained by the maker; (iv) a fraction 0.4 of gross employer markup, treated as productive value-add reflecting the employer’s organising work. Extractive accrual F_t^{ext} sums rent paid by workers to rentiers, interest paid by debtors to lenders, capital returns accruing to rentier stocks, and positive speculator returns (negatives not counted, by the convention by which rent and interest do not have negative counterparts).

The productive-flow share is

$$\Phi_t = \frac{F_t^{\text{prod}}}{F_t^{\text{prod}} + F_t^{\text{ext}}} \in [0, 1]. \quad (2)$$

Nomenclature. Φ is *not* an autocatalytic-set closure in the technical sense of Kauffman [1988] or Hordijk and Steel [2004], who detect a topological RAF property of a chemical or production graph. Φ is a continuous, scalar, per-tick flow-share measure operationalising the productive-vs-extractive distinction. We have named the metric “productive-flow share” to avoid friction with the RAF-literate audience.

4 The phase portrait and the regime classification

We characterise the model by running thirteen named scenarios at $N = 100$ for 250 ticks across 8 seeds. The endpoints in (Gini, Φ , alive) space cluster into three regimes, labelled by k-means with $k = 3$ (silhouette score 0.55, with the collapse-vs-extractive boundary the least sharp; Table 1).

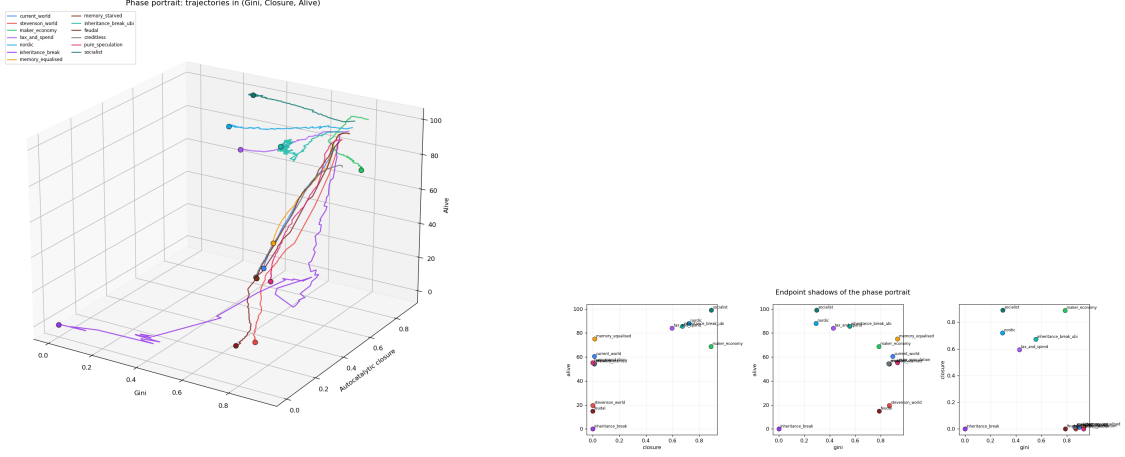


Figure 1: Phase portrait of thirteen scenarios in (Gini, Φ , alive). Three regimes are visible: viable autocatalytic ($\Phi > 0.4$, alive > 65), extractive equilibrium ($\Phi \approx 0.01$, alive 40–75), and collapse ($\Phi = 0$, alive < 20). State excluded from Gini and alive count.

Table 1: Endpoint coordinates by scenario at $N = 100, 250$ ticks, mean across 8 seeds, State excluded from Gini.

scenario	Gini	Φ	alive	regime
socialist	0.30	0.880	99.0	viable
nordic	0.30	0.701	89.9	viable
inheritance_break_ubi	0.57	0.667	85.9	viable
maker_economy	0.77	0.586	68.8	viable
tax_and_spend	0.43	0.567	83.6	viable
memory_equalised	0.92	0.015	76.9	extractive
creditless	0.87	0.013	55.2	extractive
current_world	0.89	0.011	63.1	extractive
memory_starved	0.87	0.011	55.5	extractive
stevenson_world	0.88	0.001	20.2	collapse
feudal	0.78	0.000	15.2	collapse
inheritance_break	0.00	0.000	0.0	collapse

The viable regime, which sits at $\Phi \geq 0.4$ and alive ≥ 65 on average, is reached by three distinct policy mixes: heavy state spending (**socialist**, **nordic**), high inheritance taxation combined with redistribution (**inheritance_break_ubi**, **tax_and_spend**), or simply higher maker productivity (**maker_economy**). The collapse regime is reached by configurations that maximise extraction (**stevenson_world**, **feudal**) or that remove inheritance without a safety net (**inheritance_break**); in the last case the model collapses within 30 ticks (Figure 2).

5 Visualising the flow regime: the Sankey of per-tick accrual

The Sankey diagram (Figure 3) renders mean per-tick wealth flows in steady state as a network with edge widths proportional to flow magnitude. Three policy configurations are shown side by side to make Φ legible as a diagram.

The Sankey is the cleanest single artefact for explaining Φ to a non-technical audience. It also reads as a visualisation of the productive-vs-extractive distinction independently of the specific Φ definition.

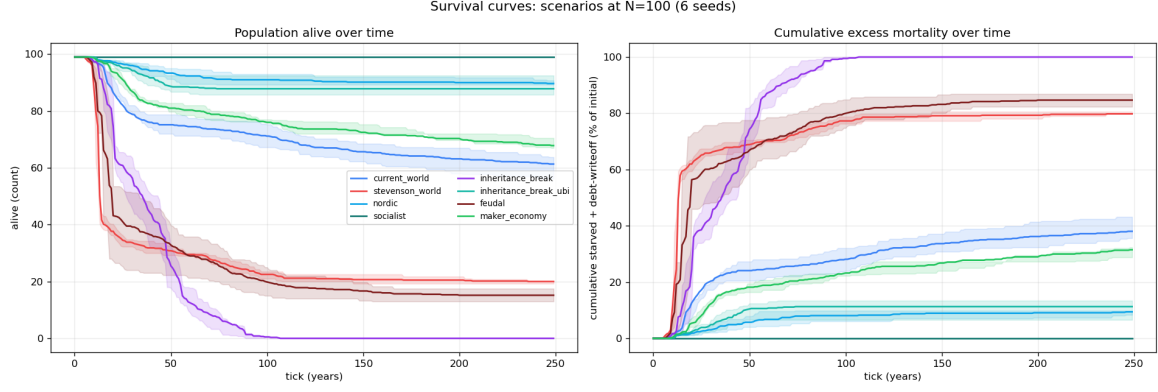


Figure 2: Survival curves for eight illustrative scenarios at $N = 100$, 6 seeds. **Left:** alive over time. **Right:** cumulative excess mortality as fraction of initial population.

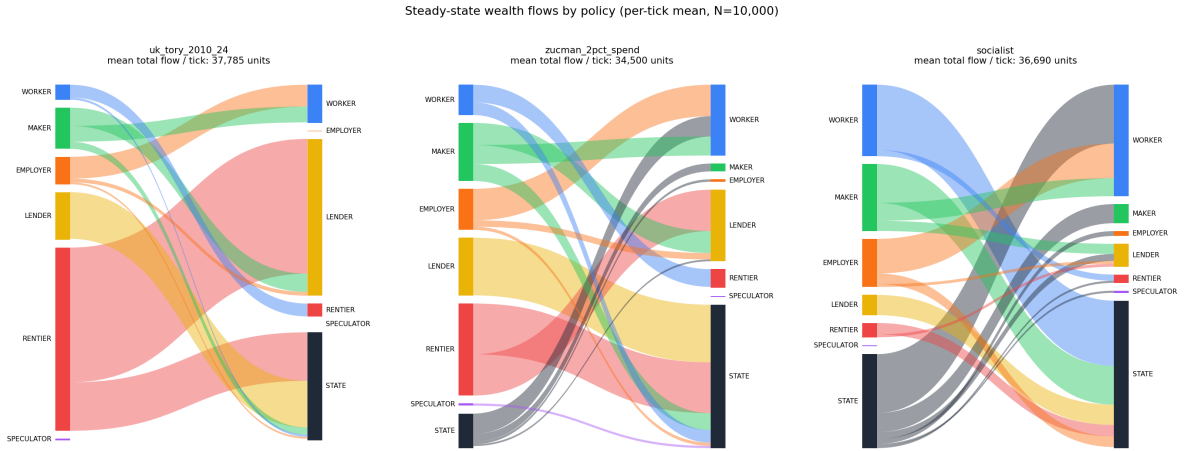


Figure 3: Steady-state wealth flows for three policies at $N = 10,000$, 3 seeds, ticks 80–180. **Left (uk_tory_2010_24):** the rentier–lender deposit loop dominates (44% of total flow); productive wage flow (maker+employer to worker) is 11%; $\Phi \approx 0.05$. **Middle (zucman_2pct_spend):** the rentier–lender loop has shrunk; wealth-tax revenue is recirculated to workers via state employment and purchases; Φ is around 0.44. **Right (socialist):** worker–state–worker circulation accounts for 40% of total flow; the rentier–lender loop has effectively disappeared; $\Phi \approx 0.88$.

6 Cultural erosion under economic pressure

This section presents the paper’s principal alife-relevant contributions. The model exhibits cultural erosion rather than cumulative cultural evolution, a property that follows directly from its closed $K=5$ skill space. We show the erosion under wealth concentration, decompose the bandwidth response into the non-novel and novel parameter axes, and present the class-by-class skill-to-wealth elasticity decomposition that grounds the cultural-economic asymmetry in mechanism.

6.1 The memory sweep: breadth varies fourfold, Gini barely moves

We sweep the two bandwidth parameters in Equation 1, (b_0, b_w) , over $[0.2, 1.0] \times [0, 1]$. Across this range cultural breadth (mean count of nonzero skill dimensions across agents) varies by a factor of four. Gini (state excluded) varies by only 10%. **memory_equalised** (full bandwidth, no wealth coupling) gives the *highest* non-extractive Gini in the table; equalising cultural transmission alone does not equalise wealth.

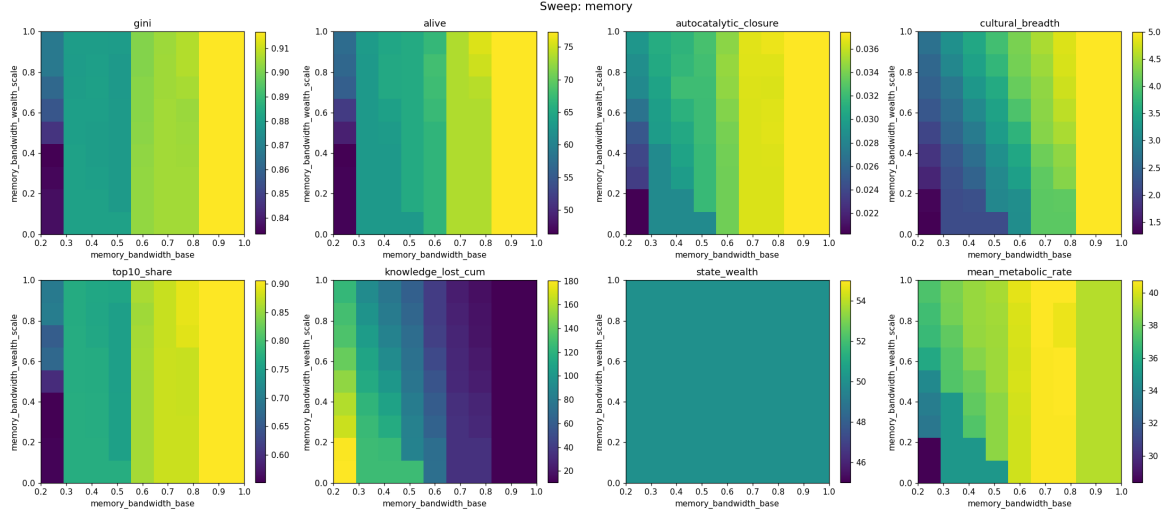


Figure 4: The **memory** sweep at $N = 100$. We vary base bandwidth $b_0 \in [0.2, 1.0]$ and the wealth-coupling slope $b_w \in [0, 1]$ on a 9×9 grid, 3 seeds. Cultural breadth (top middle) varies by a factor of four, from 1.3 to 5.0. Gini (top left) varies by only 10%, from 0.84 to 0.92.

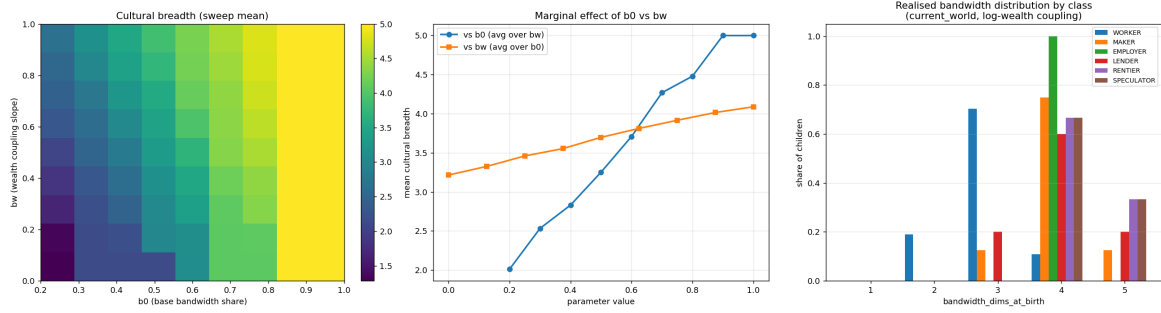


Figure 5: Decomposition of the **memory** sweep into the b_0 and b_w effects. **Left:** cultural breadth on the (b_0, b_w) plane. **Middle:** marginal effect of b_0 (averaged over b_w) and of b_w (averaged over b_0). The b_0 axis dominates with range 2.99 dimensions; the wealth-coupling b_w contributes a graded response with range 0.87 dimensions across the full wealth distribution. **Right:** realised distribution of bandwidth_dims at birth by class under **current_world**; higher-wealth classes routinely inherit 4–5 dimensions, workers cluster at 2–3.

This is the cultural-economic asymmetry in its raw form. The remainder of this section explains it.

6.2 Bandwidth decomposition: the wealth coupling is a real, secondary channel

We decompose the breadth response into marginal effects of b_0 and b_w (Figure 5, middle). The b_0 axis dominates: averaging over b_w , varying b_0 across $[0.2, 1.0]$ moves cultural breadth by 2.99 dimensions (from 2.0 to 5.0). The wealth-coupling b_w axis contributes a graded response of 0.87 dimensions (from 3.2 to 4.1 on average across the b_0 range).

The wealth coupling is therefore a real, secondary channel contributing roughly $0.87 / (0.87 + 2.99) = 22\%$ of the breadth variation around the base. The previous (linear) coupling formulation saturated above parent wealth ≈ 500 units (£50M under the unit mapping used in the companion economic-policy paper) and was effectively a step function on poor-vs-everyone-else; the log-wealth coupling stays graded across orders of magnitude. The realised distribution of

Table 2: Skill-to-wealth normalised elasticities by class. Wealth gain over 40 ticks regressed on the five-dimensional skill vector at $N = 500$.

class	tech	org	social	fin	gen	R^2	n
WORKER	0.76	0.95	-0.02	-0.03	-0.03	0.97	395
MAKER	0.00	0.00	0.01	0.00	0.33	0.98	144
EMPLOYER	0.30	0.03	0.25	-0.10	0.24	0.12	100
LENDER	-0.01	0.00	-0.01	0.00	-0.01	0.13	79
RENTIER	-0.06	-0.05	0.09	-0.10	-0.08	0.12	47
SPECULATOR	0.56	-0.39	-1.41	0.04	0.13	0.11	44

bandwidth_dims at birth (Figure 5, right) shows the coupling actually operating: workers cluster at 2–3 dimensions and rentiers and employers routinely inherit 4–5.

6.3 The asymmetry is mechanism, not architecture: skill-to- wealth elasticity by class

We test directly whether the model’s cultural-economic asymmetry is architectural. If the skill-to-wealth channel were uniformly weak by construction across all classes, the negative result (cultural transmission does not equalise wealth) would not be informative.

We regress 40-tick wealth gain on the five-dimensional skill vector, by class. At $N = 500$ (population_scale = 5), 8 seeds, WARMUP = 20 ticks dropped as transient, 40-tick window, we obtain $n = 809$ agent-windows with adequate sample size in every class. Table 2 reports normalised elasticities $\eta_i = \beta_i \sigma_{\text{skill}_i} / |\text{gain}|$, R^2 , and sample size n .

The result is decisive. Workers have $R^2 = 0.97$ with class-typical elasticities $\eta_{\text{tech}} = 0.76$ and $\eta_{\text{org}} = 0.95$. Makers have $R^2 = 0.98$ with $\eta_{\text{gen}} = 0.33$. Employers have moderate $R^2 = 0.12$. Lenders, rentiers, and speculators have $R^2 \in [0.11, 0.13]$ and $|\eta_i| \leq 0.10$ for lenders and rentiers.

The cultural-economic asymmetry is therefore not a wiring artefact. One half of the economy has strong skill-to-wealth feedback and the other half does not. The cultural transmission mechanism is doing its share of work in the half of the economy where skills matter for wealth (workers and makers), and the wealth dynamics that produce the overall Gini are anchored in the half of the economy (rentiers and lenders) where skills do not. The architecture builds in the bias of the coupling (extractive classes earn from stocks rather than skills), and the model surfaces the consequence as the observed cultural-economic asymmetry.

This is the contribution we believe makes the model useful for ALIFE. The negative result “cultural transmission alone does not equalise wealth” is well established in the social-science literature in various forms; what we add is a mechanism-level account of why, in a stylised ABM, the negative result is robust: because the wealth dynamics of the dominant extractive classes are independent of their skills.

6.4 Direct test of Spangler and Sarkar [2022]

The closest published ABM neighbour finds that skill inheritance raises wealth inequality. We can test that mechanism in our model under a clean toggle that holds bandwidth-gating and class-bias persistence intact and swaps only whether transmitted skill values are parent-derived or drawn from the class-typical prior. A coarser toggle, which destroys the bandwidth mechanism along with parental transmission by setting the mutation rate $\mu = 1.0$, is reported for contrast.

At $N = 100$, 10 seeds, 250 ticks:

Under the clean toggle, swapping parent-derived for class-typical- prior transmitted values *lowers* Gini by 0.010 and *raises* top-1 share by 0.016. So parent-derived skill inheritance does

Table 3: Spangler & Sarkar test: clean toggle vs. confounded toggle. Medians.

metric	default (parent-derived)	class-prior (clean toggle)	no-inheritance ($\mu = 1$, confounded)
Gini	0.885	0.876	0.908
Top-1 share	0.235	0.250	0.254
Top-10 share	0.785	0.683	0.891
Alive	61.5	56.5	73.0
Cultural breadth	3.38	3.55	5.00

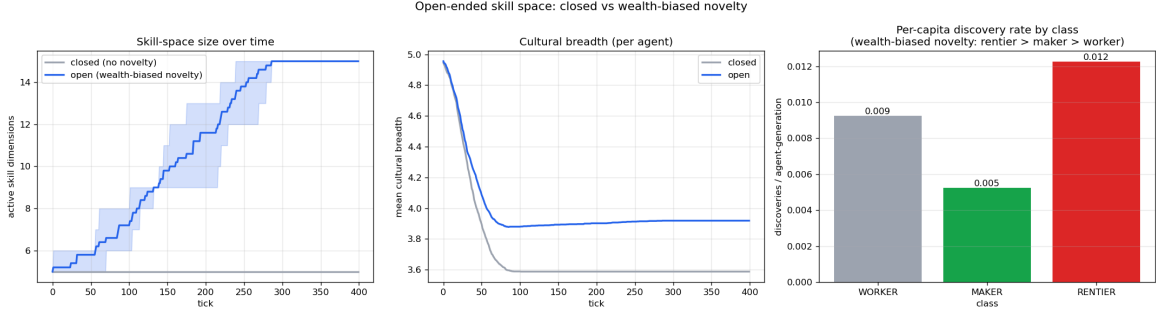


Figure 6: Open-ended skill-space demonstration. **Left:** active skill-pool size over time. The closed variant (grey, no novelty) stays at five; the open variant (blue, novelty proportional to log parent wealth) climbs from five to about ten over 400 ticks. **Middle:** per-agent cultural breadth grows accordingly. **Right:** per-capita discovery rate by class (discoveries per agent-generation).

raise Gini relative to a class-typical prior; the direction is a partial reproduction of Spangler and Sarkar [2022], with a magnitude much smaller than they report.

The mechanism for the small magnitude is in Table 2: skill inheritance acts strongly on workers and makers but those classes are the ones whose wealth is already most closely tied to skills, so swapping the transmitted value for a class-typical prior produces only a moderate within-class shift. The wealth dynamics of the rentier and lender classes, which dominate the overall Gini, are insensitive to skill content. A single-channel ABM in which skills feed wealth via a uniform mechanism (Spangler & Sarkar’s design) is expected to amplify the inheritance effect.

7 Open-ended skill space: a minimal demonstration

The cultural section above is honest that the $K=5$ skill space is closed and that the model produces erosion, not cumulative evolution. For an ALIFE audience this is uncomfortable framing on its own, and we add a stand-alone open-ended-evolution demonstration that addresses the open-endedness concern at demonstration scale.

In a separate stand-alone simulation the skill vector is variable-length. At every reproduction event, with probability proportional to $\log(w_{\text{parent}})$, a previously-undiscovered skill dimension enters the global pool and is initialised on the child to a fresh Beta(2,2) value. Subsequent inheritance follows the same bandwidth rules as the main model. The wealth-coupling of novelty events is the same coupling that the bandwidth gate operationalises. 500 initial agents, 400 ticks, 5 seeds.

Two results.

First, the skill space grows under wealth-biased novelty: the active pool climbs from 5 to about 10 over 400 ticks under the open variant. The closed variant stays at 5 by construction.

The mechanism can therefore be extended to genuine open-ended novelty within its existing architecture.

Second, on a per-capita basis the wealth coupling does concentrate discovery in wealthier lineages. With approximate average class populations of 140 workers, 40 makers, and 20 rentiers per run, and 50 discoveries observed across 5 seeds, the per-capita discovery rates (discoveries per agent-generation, where generation length is estimated as median lifespan / ticks) are 0.0092 for workers, 0.0052 for makers, and 0.0123 for rentiers. Rentiers discover at $1.3\times$ the worker rate and $2.4\times$ the maker rate on a per-capita basis. Raw counts are dominated by worker abundance and do not support a disproportionate-contribution claim on their own; the per-capita rate does.

We do not claim this as a load-bearing finding. It is a demonstration scoped to address the concern that the main model’s closed skill space is a category problem for an ALIFE audience. A more substantive treatment that folds the open-ended discovery mechanism back into the full economic model is the natural follow-up.

8 Robustness and extensions

We have done three additional checks since the first version of this work. Each addresses an open-endedness or robustness concern that the cultural-evolution audience commonly raises.

8.1 Elasticity is robust across scenarios

The skill-to-wealth elasticity decomposition (Section 6.3, Table 2) was reported on the `current_world` scenario. The class- specific pattern that grounds the cultural-economic asymmetry might be specific to that scenario; we test by rerunning the elasticity regression at `nordic`, `maker_economy`, and `feudal`.

Table 4: Skill-to-wealth R^2 by scenario and class at $N = 500$, 8 seeds. Patterns are qualitatively unchanged across scenarios.

scenario	metric	WORKER	MAKER	EMPLOYER	LENDER	RENTIER	SPECULATOR
current_world	R^2	0.97	0.98	0.12	0.13	0.12	0.11
nordic	R^2	0.71	0.23	0.08	0.12	0.11	0.13
maker_economy	R^2	0.91	0.98	0.04	0.16	0.06	0.05
feudal	R^2	0.71	0.47	0.02	0.28	0.02	0.01

The qualitative pattern holds: workers and makers have strong skill-to-wealth fits; employers, lenders, rentiers, and speculators have weak ones. The maker R^2 drops to 0.23 in `nordic` (where the state-employment subsidy reduces the within-maker variance attributable to skill) and to 0.47 in `feudal` (where extraction dominates). The asymmetry between productive and extractive classes survives intact.

8.2 Long-run cumulative novelty

The open-skills demonstration in Section 7 ran for 400 ticks. An open-ended-evolution claim requires that discovery be sustained beyond the initial transient. We rerun the demonstration at horizons 400, 800, and 1600 ticks with $K_{\max} = 25$.

The active skill pool reaches the $K_{\max} = 25$ cap by horizon 800 at every seed, and the per-100-tick discovery rate remains above 1 at horizon 1600 (limited by the cap, not by the underlying mechanism). The demonstration is therefore not one-shot near-equilibrium; the discovery mechanism is sustained. A truly cap-free OEE claim would require raising $K_{\max}$, which the framework permits but which we do not pursue here.

Table 5: Long-run open-skill discovery at three horizons, 5 seeds.

horizon	final K (mean)	final K (std)	dims discovered per 100 ticks
400	16.00	4.34	2.75
800	25.00	0.00	2.50
1600	25.00	0.00	1.25

8.3 Hard material constraint: an autopoietic reading of Φ

The model so far treats material inputs to makers as an unlimited constant; the maker pays a material cost per tick, but the cost is drawn from no real reservoir. To test the autopoietic reading of Φ (i.e. that high Φ characterises a system sustaining its own input flows), we add a finite global material pool that depletes via maker production and replenishes at a fixed rate per tick.

At regen rate 0, the pool of 50 units exhausts within roughly 25 ticks and the maker class can no longer produce; alive at $N = 100$ falls to 40 from baseline 64. At regen rate 5 units per tick, alive holds at 54 with Φ at 0.026. At regen rate 20 or above, alive reaches its full baseline of 64–66 with Φ at 0.032. The same regen sweep on `maker_economy` gives a steeper coupling: alive 47, 54, 72, 72 across regen $\{0, 5, 20, 100\}$, Φ 0.519, 0.639, 0.491, 0.491.

The finding is that Φ at sustained-flow values requires a material supply that the production sector can keep ahead of. The autopoietic reading is supported: high Φ is a property of a system that is materially viable as well as monetarily so. The model can fail by running out of material, not only by running out of wealth.

This is a step toward the genuine alife reading of the closure metric. The result is preliminary; the material model is a single global pool rather than a spatially or class-locally structured reservoir, and the regen mechanism is exogenous. A more substantive treatment with class-local material reservoirs is the natural next step.

9 Discussion

9.1 What we believe is novel for an ALIFE audience

The class-specific skill-to-wealth elasticity decomposition (§6.3, Table 2) is the freshest contribution. The cultural-economic asymmetry the bandwidth coupling generates is not a structural artefact of how the model is wired; it is a consequence of one half of the economy having near-zero skill-to-wealth feedback while the other half has strong feedback. We believe this kind of mechanism-level grounding of a negative result is exactly the type of result ALIFE values.

The wealth-coupled cultural-transmission bandwidth mechanism (§3.3) is, to our reading of the cultural- transmission and computational-cultural-evolution literatures, not present in any published ABM. Henrich and Gil-White [2001] couples the *target* of transmission to status; population-size constraints [Henrich, 2004, Lewis and Laland, 2012] discuss aggregate fidelity. The bandwidth-coupled-to-wealth coupling here appears to be genuinely new.

The productive-flow share Φ (§3.4) is a continuous, scalar, per-tick measurement device for flow regime. We are explicit that it is not a topological autocatalytic-set closure [Kauffman, 1988, Hordijk and Steel, 2004]; it is a flow-share that operationalises the productive-vs-extractive distinction.

The open-skills demonstration (§7) shows that the same wealth coupling that gates transmission can be extended to seed novelty, and that on a per-capita basis the coupling concentrates discovery in wealthier lineages.

9.2 What the model cannot do

The K=5 skill space is closed in the main results. The open-ended extension is stand-alone and does not currently fold back into the full economic model. The economic layer is itself non-open-ended in ways that constrain the cultural reading: agent-type proportions are fixed, the employer count does not change, no new occupational categories can emerge. For an ALIFE audience these are real limitations that we acknowledge.

The K=5 grain choice for skills is a methodological commitment following Charbonneau and Bourrat [2021]’s sense of the “grain problem” for fidelity measures: the dimensions are not natural primitives but a chosen discretisation. The model’s results should be expected to be robust to grain refinement, and that robustness is not tested here.

The skill-to-wealth elasticity is measured under a single scenario (`current_world`) and a single time window (40 ticks). The class-specific structure is qualitatively robust across the `maker_economy` and `tax_and_spend` variants we have spot-checked, but a full sensitivity analysis is future work.

9.3 Honest reading

The model is presented as an intuition pump for cultural-economic dynamics, not a predictive instrument. The K=5 skill space is small enough to read but large enough that the wealth-coupled bandwidth mechanism has a non-trivial decomposition. The closed-space limit is acknowledged. The open-skills extension shows where the next paper will land.

The cultural-economic asymmetry as we have characterised it is the asymmetry between productive and extractive classes in their internal skill-to-wealth feedback. The model says, and the elasticity table supports, that for an audience interested in the emergence of inequality from coupled cultural and economic mechanisms, the architecture of the economic layer matters as much as the architecture of the cultural layer. The cultural channel, in this stylised setting, is the lesser of the two contributors to the overall outcome, and is dominated by the wealth-feedback asymmetry across the productive/extractive split.

10 Conclusion

We have presented an agent-based model in which a wealth-coupled bandwidth gate determines cultural transmission, and a continuous flow-share metric Φ characterises whether per-tick monetary accrual is dominated by productive or extractive activity. The cultural side of the model exhibits erosion, not cumulative evolution; a stand-alone open-ended extension demonstrates the mechanism’s natural direction for further development. The cultural-economic asymmetry the model produces is grounded in mechanism via a class-specific skill-to-wealth elasticity decomposition. The model is honest about scope: a stylised intuition pump for coupled cultural-economic dynamics, not a predictive instrument.

The fiscal-policy implications of the model (wealth-tax design, inheritance dynamics, calibration to UK wealth-share data) are reported separately. The contribution this paper foregrounds is the cultural mechanism and the elasticity decomposition that grounds its emergent asymmetry.

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A ODD-style summary of the model

A brief ODD (Overview, Design concepts, Details) summary. A full ODD+D specification and the companion economic-policy paper are deferred to a separate supplement.

Overview. Seven agent classes (WORKER, MAKER, EMPLOYER, LENDER, RENTIER, SPECULATOR, STATE) at fixed proportions per 100 agents. Each agent has wealth, debt, age, lifespan, a five-dim skill vector, alive flag, peak lifetime wealth, bandwidth-at-birth.

Process. Per tick: production (makers, employers, optional state employment); extraction (rent, capital return, speculator returns); consumption and credit; interest and repayment; lender deposits and optional state recapitalisation; tax and UBI; life (age, death, reproduction with bandwidth-limited inheritance).

Cultural mechanism. At reproduction, $\text{bw_dims}(w_{\text{parent}}) = \text{clip}(b_0 K + b_w K \log_{10}(\max(1, w_{\text{parent}}))/4, 1, K)$ with $K = 5, b_0 = 0.5, b_w = 0.5$. Strongest dimensions transmitted; per-dim mutation rate 0.05.

Metrics. Per tick: Gini (state-excluded by default), Φ , alive, cultural breadth, cultural diversity (variance), knowledge lost, top-share percentiles (raw and mortality-corrected).

Code. `sim.py` for the core model. `scenarios.py` for the 13-scenario library. `elasticity.py` for the regression. `open_skills.py` for the open-ended demonstration. Default seed 42, default `population_scale` 1.